Thumma Tejaswi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.25
SR PRIME SCHOOL HANAMKONDA WARANGAL TELANGANA	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2024	96.00%
NATIONAL (EM) HIGH SCHOOL, HANAMKONDA, HANUMAKONDA DISTRICT	Class X (Board of Secondary Education, Telangana State)	2022	10.00%

PROJECTS

Arbitrary Precision Arithmetic Library in Java

- Developed a custom arithmetic Java library for integer and floating-point arithmetic with **unlimited precision** using **String-based computation**. Implemented addition, subtraction, multiplication, and division with full-sign handling and precise decimal alignment.
- Achieved high numerical accuracy by supporting up to **30 decimal places** and eliminating **overflow limitations** of native data types.

Exploiting Floating Point Units for Integer Workloads

- Exploited idle Floating Point Units to increase **instruction-level parallelism** in integer-heavy workloads by redesigning benchmarks with mixed ALU-FPU execution and optimized integer-to-float operation ratios.
- Implemented mixed integer-float versions of dot product, matrix multiplication, factorial, power, and prime-checking using **loop unrolling** and ratio tuning (2:1 to 5:1) with **-O3 compiler optimizations**.
- Automated benchmarking using Python scripts to run each workload across multiple inputs, compute median execution times over 10 runs, and generate performance comparison plots across integer and mixed implementations.
- Profiled and compared results across 2 architectures, Apple M3 (ARM) and Raspberry Pi 5 (Cortex-A76) using perf stat and clock(), achieving **speedups ranging from 1.2x to 3.4x** depending on workload and architecture.

xv6 Operating System Extensions and Kernel Subsystems

- Extended the xv6 RISC-V operating system by implementing custom system calls, process accounting, and concurrent parent-child management inside the kernel.
- Designed and implemented a **System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ)** scheduler with priority boosting, syscall-aware demotion, and detailed per-process scheduling statistics.
- Developed a scheduler-aware **virtual memory** subsystem using **Clock page replacement**, **lazy allocation**, swap management, and per-process memory statistics tracking.
- Implemented disk-backed swap management with **FCFS** and **SSTF** disk scheduling policies, along with **RAID-0**, **RAID-1**, and **RAID-5** based storage simulation inside xv6.

Bitcoin-Style Blockchain Implementation

- Built a **Bitcoin-style blockchain** with wallets, transactions, **Proof-of-Work mining**, and a **UTXO ledger** to manage ownership and spending of coins.
- Designed **layered double-spend prevention** by rejecting conflicting transactions in the **mempool**, detecting duplicate UTXO usage within a block, and validating spends against the confirmed UTXO set.
- Secured transactions using **ECDSA** over **SECP256k1**, where every transaction input is digitally signed with the sender's private key and verified using the corresponding public key before acceptance.
- Implemented **Merkle tree construction**, coinbase transactions, reward-halving logic, transaction fees, change outputs, and blockchain validation.

AI Resume Analyzer

- Built a full-stack AI-integrated application using **prompt engineering with Google's Gemini API** to extract and analyze resume content against a job description, returning a match score, missing keywords, red flags, and suggested improvements via **schema-constrained structured JSON output**.
- Architected the backend into independently testable modules (document parsing, prompt building, API integration) with asynchronous FastAPI endpoints handling file uploads, type validation, and error responses.

RELEVANT COURSES

- **Computer Science:** Data Structures, Algorithms, Operating Systems, Database Management Systems, Theory of Computation, Computer Architecture, Software Development Fundamentals, Introduction to Programming, Compilers I, Introduction to Artificial Intelligence.

- **Mathematics:** Linear Algebra, Transform Techniques, Discrete Mathematics, Calculus, Differential Equations, Probability and Random Variables, Complex Variables, Introduction to statistics.

SKILLS

- **Programming:** C/C++, Python, Java, RISC-V assembly
- **Web Development:** HTML, CSS, JavaScript
- **Databases & Tools:** PostgreSQL, Git, GitHub, Jupyter Notebook, Google Colab
- **Systems & Technical skills:** xv6, LaTeX

POSITIONS OF RESPONSIBILITY

- **Head**, Blockspace (Blockchain and Crypto club), IIT Hyderabad (2026 - present)
- **Core**, Blockspace (Blockchain and Crypto club), IIT Hyderabad (2025-2026)
- **Web Coordinator**, Milan (General Championship), IIT Hyderabad (2025-2026)
- **UG Mentor**, Sunshine (The Counselling Cell), IIT Hyderabad (2025-2026)